

SKARPO, Sweden

WMO: 024870

Lat: 59.345N

Lon: 18.740E

Elev: 22

StdP: 101.06

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.2	-9.6	-14.5	1.1	-11.2	-12.0	1.3	-8.3	12.1	-1.2	11.0	-1.6	3.3	290	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	24.9	18.3	23.3	17.4	21.8	16.6	19.7	22.9	18.7	21.5	17.7	20.6	3.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.5	21.2	17.5	12.6	20.1	16.5	11.8	19.2	56.6	23.0	53.0	21.6	50.1	20.6	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.1	8.0	DB	-14.8	27.0	4.8	1.6	-18.3	28.1	-21.1	29.1	-23.8	29.9	-27.2	31.1	(4)
(5)			WB	-15.1	20.6	4.5	1.4	-18.3	21.6	-20.9	22.5	-23.5	23.3	-26.8	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-1.3	-1.7	1.0	5.1	10.2	14.9	18.0	17.3	13.2	8.1	4.0	0.9		
	DBStd	7.57	3.95	4.21	3.69	2.90	3.02	2.54	2.46	2.31	2.43	3.10	3.04	3.88		
	HDD10.0	1681	350	328	279	149	34	1	0	0	3	73	181	281		
	HDD18.3	3989	608	562	538	397	252	106	38	47	154	316	431	539		
	CDD10.0	778	0	0	0	2	41	147	247	226	100	16	0	0		
	CDD18.3	44	0	0	0	0	0	3	26	15	1	0	0	0		
	CDH23.3	134	0	0	0	0	1	11	86	37	0	0	0	0		
	CDH26.7	9	0	0	0	0	0	0	7	3	0	0	0	0		
(14) Wind	WSAvg	3.9	4.3	4.1	3.9	3.5	3.5	3.5	3.4	3.3	3.8	4.0	4.5	4.4		
(15) Precipitation	PrecAvg	559	41	34	28	30	36	57	54	62	55	54	52	55		
	PrecMax	665	86	63	53	77	70	108	106	144	151	102	100	123		
	PrecMin	394	7	8	1	7	11	13	13	2	10	26	13	11		
	PrecStd	64	20	13	12	21	16	28	27	38	39	19	25	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	7.9	12.0	15.9	21.7	24.9	27.4	26.6	21.8	15.6	10.7	9.7		
		MCWB	5.5	4.9	7.2	9.7	13.1	16.4	18.9	19.7	16.5	12.9	8.9	7.9		
	2%	DB	5.6	5.7	9.5	13.4	18.7	22.6	25.6	24.2	19.3	14.3	9.6	7.9		
		MCWB	4.4	3.6	5.5	8.5	12.3	15.0	18.6	18.2	15.2	12.4	8.5	6.6		
	5%	DB	4.7	4.4	7.4	11.5	16.9	21.1	24.0	22.5	17.9	13.3	8.9	6.9		
		MCWB	3.7	3.1	4.5	7.4	11.5	14.6	18.0	17.7	14.6	11.7	8.0	5.9		
	10%	DB	3.7	3.4	5.6	9.7	15.1	19.5	22.5	21.1	16.7	12.2	8.1	5.8		
		MCWB	2.8	2.3	3.3	6.3	10.5	14.2	17.4	16.8	13.9	10.9	7.2	4.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.4	7.6	10.6	14.5	17.9	21.7	21.6	17.2	13.7	9.5	8.5		
		MCDB	6.7	7.1	11.2	15.1	19.2	21.7	23.9	24.6	20.2	14.5	10.0	9.5		
	2%	WB	4.7	4.1	6.0	9.0	13.2	16.4	20.2	19.4	16.0	12.8	8.7	6.9		
		MCDB	5.3	5.1	9.0	12.8	17.5	20.6	23.6	22.0	18.2	13.8	9.3	7.6		
	5%	WB	3.8	3.2	4.8	7.9	12.0	15.5	19.2	18.5	15.2	11.9	8.1	6.1		
		MCDB	4.5	4.2	7.1	10.9	15.6	19.3	22.3	21.1	17.1	13.1	8.8	6.7		
	10%	WB	2.9	2.4	3.8	6.7	11.0	14.8	18.2	17.7	14.5	11.0	7.4	5.0		
		MCDB	3.7	3.4	5.2	9.2	14.4	18.2	21.2	20.2	16.2	12.0	8.0	5.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	4.4	5.3	6.6	6.9	6.9	6.3	5.9	4.9	4.0	3.2	3.4		
		MCDBR	4.0	4.8	7.7	9.5	9.7	9.5	8.4	7.5	6.3	4.6	3.7	3.7		
	5% WB	MCWBR	3.8	3.7	4.9	5.5	4.9	4.2	3.2	3.2	3.3	3.2	3.4	3.4		
		MCWBR	3.7	4.4	7.1	9.0	8.4	8.1	6.9	6.4	5.5	4.3	3.5	3.5		
(40) Clear-Sky Solar Irradiance	taub		0.287	0.322	0.339	0.376	0.371	0.363	0.391	0.382	0.350	0.338	0.312	0.271		
		taud	2.227	2.298	2.355	2.326	2.412	2.475	2.416	2.434	2.505	2.414	2.248	2.142		
	Ebn at Noon		512	687	797	825	859	872	834	814	786	666	467	379		
		Edh at Noon	44	68	87	106	104	99	103	95	75	61	43	32		
(44) All-Sky Solar Radiation	RadAvg		0.28	0.81	2.32	4.06	5.47	6.06	5.65	4.45	2.84	1.22	0.38	0.19		
	RadStd		0.04	0.15	0.23	0.37	0.44	0.33	0.51	0.35	0.30	0.20	0.09	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		+0.34	N/A	N/A	N/A	N/A	N/A	-106	-129	N/A	N/A

Nomenclature: See separate page